

Design Calculation Sheet — VBF-T7R1NOFORCE-CALC-01

Case t7_r1_noforce — 2026-08-25. PDF summary of calc.xlsx (5 sheets: inputs → capacity/energy → energy density → N/P & mass → process parameters).

Property	Value	Source
Charge capacity	5.0065 Ah	r5_slim_1c_dfn.json
Energy	17.8042 Wh	r5_slim_energy.json
Volumetric	491.471 Wh/kg = 17.8042 / 0.0362264	r5_slim_energy.json
Gravimetric	928.06 Wh/L = 17.8042 / 0.019184	r5_slim_energy.json
Contract	36.2264 g (electrolyte excluded)	r5_slim_energy.json
Capacity density	666,017 Ah/m3 = 33133 x 0.75 x 96485/3600	Chen2020
Capacity density	1,124,723 Ah/m3 = 63104 x 0.665 x 96485/3600	Chen2020
	0.667 = (666017 x 85.2e-6) / (1124723 x 75.6e-6); usable-window 1.00	formula
Area density	163.99 g/m2 = 75.6e-6 x 0.665 x 3262	calc-energy
Area density	105.88 g/m2 = 85.2e-6 x 0.75 x 1657	calc-energy
Reaction pos	2.169 g/cm3 = 3262 x 0.665 / 1000	formula
Reaction neg	1.243 g/cm3 = 1657 x 0.75 / 1000	formula
Electrolyte fill	6.33 g = 5.27 cm3 x 1.2 g/cm3 x 1.0	formula (lit density)